

AN EFFICIENT IMAGE COMPRESSION METHOD BASED ON NEURAL NETWORK: AN OVERFITTING APPROACH

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ABSTRACT

Over the past decade, nonlinear image compression techniques based on neural networks have been rapidly developed to achieve more efficient storage and transmission of images compared with conventional linear techniques. A typical nonlinear technique is implemented as a neural network trained on a vast set of images, and the latent representation of a target image is transmitted. In contrast to the previous nonlinear techniques, we propose a new image compression method in which a neural network model is trained exclusively on a single target image, rather than a set of images. Such an *overfitting strategy* enables us to embed *fine* image features in not only the latent representation but also the network parameters, which helps reduce the reconstruction error against the target image. The effectiveness of our method is validated through a comparison with conventional image compression techniques in terms of a rate-distortion criterion.

Index Terms— Image compression, neural network, overfitting, rate-distortion performance, parameter visualization.

1. INTRODUCTION

1.1. Motivation

Image compression is a fundamental task in image processing and is used in a variety of applications, such as storage and transmission. Modern image compression techniques can be classified into two types: linear and nonlinear. The linear types utilize the linear transformation of a target image, e.g., the discrete cosine and wavelet transformations, to exploit the spatial correlation of pixel values [1, 2]. For the same purpose, the nonlinear types rely on the nonlinearity of neural networks (NNs) [3]. Over the past decade, nonlinear NN-based techniques have attracted considerable attention because they have potential to be comparable to the conventional linear ones in a rate-distortion (RD) sense. In the present work, we investigate how to improve the RD performance of such nonlinear techniques.

We start by briefly summarizing recently developed image compression techniques using NNs. Ballé et al. [4] proposed an image compression technique based on an autoencoder [5], where the network parameters were optimized against a vast set of training images so as to minimize the reconstruction

errors and entropy values for the corresponding latent representations. They then transmitted the latent representation of a single target image after quantization and entropy coding. This technique has been improved by reformulating the loss function in [6–9]. Toderici et al. [10] proposed another technique based on a recurrent NN [11] in which they optimized the network parameters for many images (as in [4, 6–9]) and transmitted residual information in addition to the latent representation. A successor of this technique has been proposed by the same research group [12]. Other works can be found in the literature (e.g., see [13–16]).

The NN-based image compression techniques proposed thus far have been trained against a vast set of training images to achieve a certain level of compression performance for arbitrary input images. However, such an approach often leads to a large error between the target and decoded images under the limited potential of the network, e.g., small numbers of hidden layers as well as network parameters. In other words, small-scale NNs cannot precisely represent fine features of the target image such as edges and textures. This undesirable behavior degrades the compression performance, especially in higher bitrates (as reported in [6]). Although using a deep residual network architecture [17] (as in [7]) is one solution to circumvent this difficulty, the training procedure becomes harder due to the requirement of huge memory space for storing the network parameters.

In light of the above, we propose a new image compression method implemented with a small-scale NN. Our strategy is to extremely reduce the number of training images: a vast set of images is replaced with a single target image. Thanks to this replacement, the NN model in our method is fitted only to the single target image, rather than to a set of images as done in the previous techniques. As a result, the trained model is optimized to represent fine features of the target image, which leads to fewer reconstruction errors and a better compression performance.

1.2. Contributions

As will be detailed in Section 2, our method requires a transmission of *network parameters* in addition to the latent representation. We should thus optimize both the network parameters and the latent representation through the training and

build a compression algorithm for them. The main contributions of this paper are twofold:

- A formulation of the loss function including the entropy of the network parameters.
- Construction of an encoding method for the network parameters.

Note that He et al. [18] have proposed a video compression technique based on the overfitting of an NN model, which is somewhat similar to our approach. They first transformed a target video into a bitstream by using a conventional linear encoder, e.g., high efficiency video coding [19], and then trained the network parameters using only the obtained bitstream to remove underlying compression noise. This technique requires both the linear encoder and the NN. In contrast, our method is performed with an NN alone, which is the difference between our method and [18].

1.3. Notations

Throughout this paper, let x and z be the target and decoded images with $I \times J$ pixels and K color channels, respectively, and y be the latent representation of x . We denote the non-linear analysis (encoding) and synthesis (decoding) operators equipped with network parameters ϕ and ψ by f_ϕ and g_ψ , respectively. We represent a quantization operation by Q and describe x and z as follows:

$$\underbrace{y = f_\phi(x)}_{\text{analysis}} \rightarrow \underbrace{y' = Q(y)}_{\text{quantization}} \rightarrow \underbrace{z = g_\psi(y')}_{{\text{synthesis}}}. \quad (1)$$

2. OUR METHOD

2.1. Loss Function

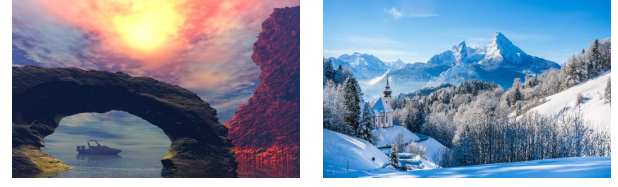
We first describe the loss function for optimizing (1) in our method. Because we fit an NN model only to the single target image x in our scenario, ψ is optimized to this image, rather than to a set of images. Let $\psi' = Q(\psi)$ be a quantized version of ψ . We have to transmit ψ' in addition to y' to perform the synthesis operation $z = g_\psi(y')$ at the decoder side. For this reason, we formulate a loss function as

$$\ell(\phi, \psi) = \mathbb{E} \left[D(x, z) + \lambda_{y'} \cdot R(y') + \lambda_{\psi'} \cdot R(\psi') \right], \quad (2)$$

where $D(x, z) = \frac{1}{I \cdot J \cdot K} \cdot \|x - z\|_2^2$ is the mean squared error (MSE), $R(y')$ and $R(\psi')$ are the entropy of y' and ψ' , respectively, and $\lambda_{y'}$ and $\lambda_{\psi'}$ are positive parameters. Following [4], we compute the expected entropy terms by estimating the probability density functions of $y + u_y$ and $\psi + u_\psi$ (u_y and u_ψ are uniform noises) from the continuous relaxations of the histograms for y' and ψ' , respectively. The actual parameter values $\lambda_{y'}$ and $\lambda_{\psi'}$ are given empirically in Section 3.2.

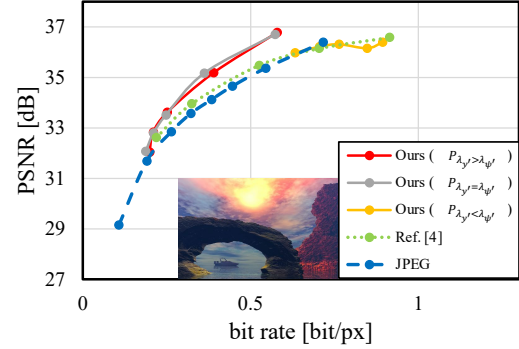
Remark 1. In the previous techniques [4, 6–10, 12–16], a loss function of the form

$$\ell(\phi, \psi) = \sum_{x \in \mathcal{X}} \mathbb{E} \left[D(x, z) + \lambda_{y'} \cdot R(y') \right] \quad (3)$$

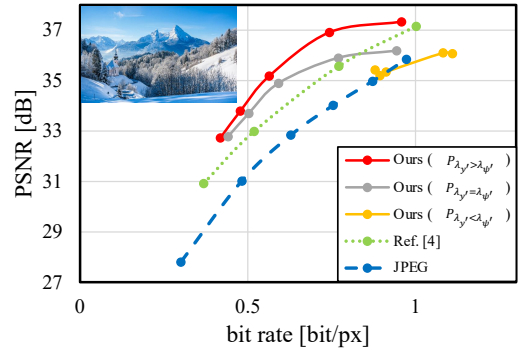


(a) (b)

Fig. 1. Target images ($7680 \times 4320 \times 3$).



(a)



(b)

Fig. 2. Rate-distortion performances.

is minimized to optimize ϕ and ψ over a vast set of training images $\mathcal{X}$. To reduce the loss, the network should be of sufficiently large-scale; a small-scale NN cannot be generalized for a large amount of training images. Moreover, an NN trained over many images is more likely to neglect fine edges and textures specific to each image. In contrast, we adopt a small-scale NN but embed the features of a single target image to both ψ' and y' . Our method can thus achieve a better reconstruction quality for the fine details.

2.2. Coding of y and ψ

Various quantization and entropy coding techniques for a latent representation y have been studied extensively in previous works [4, 6–10, 12–16]. Following [4], we encode y in (2) by using ordinary scalar quantization and context-based adaptive binary arithmetic coding [20].

Table 1. Network architecture in [4].

Layer	Ker. size	Stride	#channels(in/out)	Acc. strides(in/out)	Input
conv-1 + GDN-1	9	4	3/128	1/4	Input image
conv-2 + GDN-2	5	2	128/128	4/8	conv-1 + GDN-1
conv-3	5	2	128/128	8/16	conv-2 + GDN-2
deconv-1 + IGDN-1	5	2	128/128	16/8	conv-3
deconv-2 + IGDN-2	5	2	128/128	8/4	deconv-1 + IGDN-1
deconv-3	9	4	128/3	4/1	deconv-2 + IGDN-2

Table 2. Network architecture in our method.

Layer	Ker. size	Stride	#channels(in/out)	Acc. strides(in/out)	Input
conv-1' + GDN-1'	33	16	3/72	1/16	Input image
deconv-1' + IGDN-1'	33	16	72/3	16/1	conv-1' + GDN-1'

We hereafter mention the quantization and entropy coding techniques for the network parameter ψ . Let ψ_d be network parameters of the d -th hidden layer at the decoder, $\mathbf{W}_d$ and $\mathbf{b}_d$ be the weights and biases that constitute ψ_d , respectively, and $\bar{w}_d$ and $\bar{b}_d$ represent the mean values of $|\mathbf{W}_d|$ and $|\mathbf{b}_d|$ for each layer d , respectively. For $\mathbf{W}_d$ and $\mathbf{b}_d$, we first determine their effective numbers of binary digits as

$$N(\bar{w}_d) = \text{round}\left(\log_2(C_{\bar{w}_d} \cdot \bar{w}_d)\right), \quad (0 < C_{\bar{w}_d} < 1) \quad (4)$$

and

$$N(\bar{b}_d) = \text{round}\left(\log_2(C_{\bar{b}_d} \cdot \bar{b}_d)\right), \quad (0 < C_{\bar{b}_d} < 1), \quad (5)$$

where the parameters $C_{\bar{w}_d}$ and $C_{\bar{b}_d}$ are used to control the probability of the symbols '0' and '1' in the binary digits. We then apply the quantization and dequantization to $\mathbf{W}_d$ and $\mathbf{b}_d$ by the step sizes $2^{N(\bar{w}_d)}$ and $2^{N(\bar{b}_d)}$, respectively. Note that the range of the noise term $\mathbf{u}_{\psi}$ depends on these step sizes. Since the 32-bit floating-point representations of the dequantized values have many zeros, we finally compress them by using the classical but efficient Lempel-Ziv-Markov chain algorithm [21] for each layer d .

Remark 2. The size of the latent representation $\mathbf{y}'$ depends on that of the target image $I \times J$. The bit amount of $\mathbf{y}'$ thus becomes significantly large for a high-resolution image. On the other hand, the number of parameters $\mathbf{W}_d$ and $\mathbf{b}_d$ are invariant for the image size. This useful characteristic enables us to reduce the bit amount of $\mathbf{y}'$ for a high-resolution image by using a large $\lambda_{\mathbf{y}'}$ and small $\lambda_{\psi'}$.

3. EXPERIMENTAL RESULTS

3.1. Implementation and Training

We implemented our framework on an autoencoder-based method [4]. We removed two hidden layers from the original network in [4], as we aimed to validate the effectiveness of our method with a less complex network structure. Tables 1

Table 3. Parameter values in our method.

Parameter set	$(\lambda_{\mathbf{y}'}, \lambda_{\psi'})$
$P_{\lambda_{\mathbf{y}'} > \lambda_{\psi'}}$	(100, 10), (250, 25), (500, 50), (750, 75), (1000, 100)
$P_{\lambda_{\mathbf{y}'} = \lambda_{\psi'}}$	(100, 100), (250, 250), (500, 500), (750, 750), (1000, 1000)
$P_{\lambda_{\mathbf{y}'} < \lambda_{\psi'}}$	(10, 100), (25, 250), (50, 500), (75, 750), (100, 1000)

and 2 list the network structures of [4]¹ and our method, respectively, where **conv** denotes a convolutional layer with a downsampling operation, **deconv** is a deconvolution layer with upsampling (zero-filling), and **GDN** and **IGDN** [22] are an activation function and its approximate inverse. Note that the spatial size of the latent representation was determined by that of the target image, $I \times J$. However, for fair comparison, we ensured that the latent representations of both methods had the same spatial sizes for the same target image.

We trained the original method [4] using about 6,000 images as $\mathcal{X}$ in (3). In contrast, we trained our NN models individually for target images, which made each NN overfit to a specific target image. Figure 1 shows two target color images with $I \times J \times K = 7680 \times 4320 \times 3$. The loss function (2) was minimized by the Adam optimizer [23] with the learning rate 10^{-5} . Our NN models for the target images (a) and (b) were trained with 100K and 70K epochs, respectively. Table 3 summarizes the actual values for $\lambda_{\mathbf{y}'}$ and $\lambda_{\psi'}$, which significantly affected the result. The parameters $C_{\bar{w}_1}$ and $C_{\bar{b}_1}$ were set to 0.01.

3.2. Evaluation

We compared our method with the original autoencoder-based technique [4] and JPEG. For each method, we quantitatively evaluated the image quality by the peak signal-to-noise ratio (PSNR) [dB] between $\mathbf{x}$ and $\mathbf{z}$, and the coding efficiency

¹We used a Tensorflow implementation from <https://github.com/tensorflow/compression>.

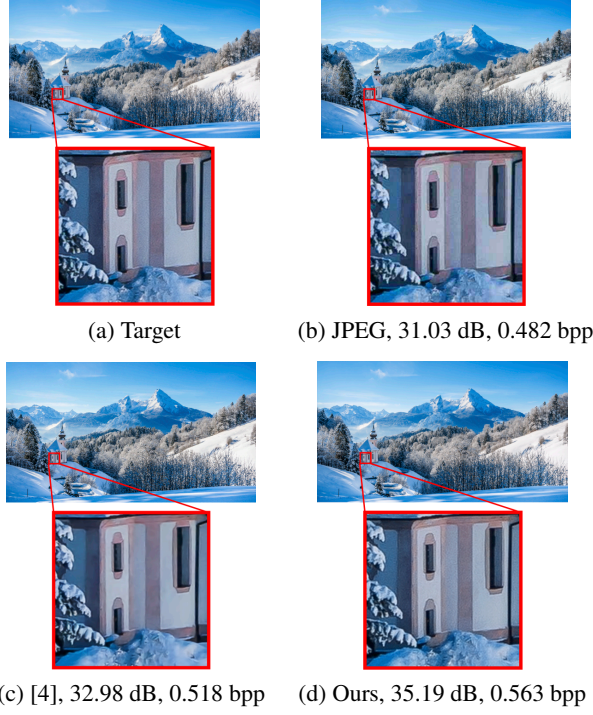


Fig. 3. Decoded images.

by bits per pixel [bpp].

Figure 2 shows the RD curves of each method, where (a) and (b) correspond to the target 8K images in Fig. 1. We can see from these results that our method with appropriate $\lambda_{y'}$ and $\lambda_{\psi'}$ outperformed both JPEG and [4]. Note that the better performance of our method was achieved with the simplified network structure, indicating that a deeper network is not required to improve the RD performance thanks to the overfitting strategy adopted in our method. We should also point out that the performances of our method with small $\lambda_{y'}$ and large $\lambda_{\psi'}$ were slightly degraded compared to [4]. This is because the bit amount of y' depends on (is proportional to) the spatial resolution $I \times J$ and becomes large with small $\lambda_{y'}$. However, this degradation can be avoided by setting $\lambda_{y'} \geq \lambda_{\psi'}$.

Figure 3 shows examples of images decoded by (b) JPEG, (c) [4], and (d) our method. As we can see, textures and edges in (b) and (c) were blurred and oversmoothed, while our result (d) obviously preserved fine image features. These results demonstrate the advantage of our overfitting strategy in terms of the RD performance and qualitative image quality.

3.3. Visualization of W_d and y'

We then investigate why the fine image features can be represented efficiently in our method. Figure 4(a) and (b) depicts kernel weights from **deconv-1** in [4] and **deconv-1'** in our method, where we chose the kernels with the maximum spatial variances over the input and output channels in these layers. Figure 4(c) and (d) are the frequency responses of (a) and (b), respectively. From these results, the weights in Fig. 4(a) can be interpreted as a low-pass filter while our

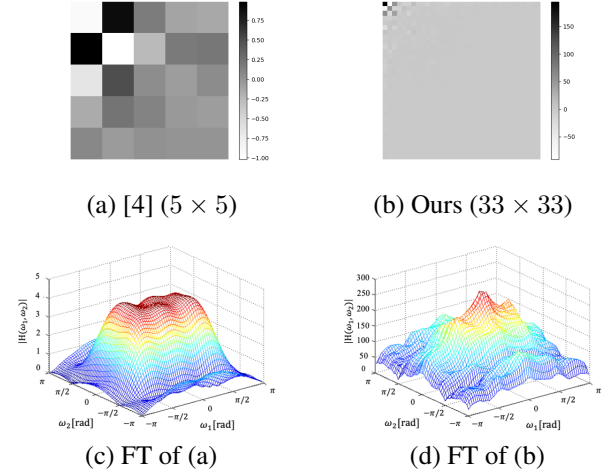


Fig. 4. Visualization of W'_d and its Fourier transform (FT).

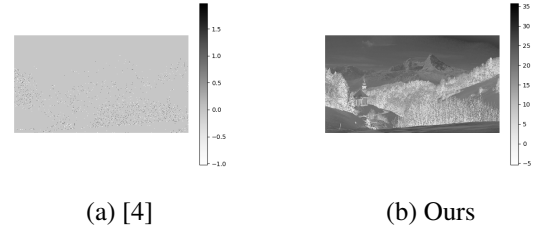


Fig. 5. Visualization of y' .

weights in Fig. 4(b) constituted a filter that passes through high-frequency components as well. Figure 5 shows latent representations y' of [4] and our method corresponding to Fig. 4(a) and (b), respectively, where (b) includes more details than (a), such as edges and textures. We conclude that our overfitting strategy enabled us to embed instance-specific details and subtle features into the latent representation and network weights, resulting in a more accurate reconstruction of the target image.

4. CONCLUSION

In this paper, we proposed a nonlinear image compression method based on NNs. We simplified the network structure and overfitted the NN model to only a single target image. We also proposed a compression method for network parameters based on quantization and entropy coding. In comparison with conventional techniques, we showed the advantages of our method in terms of an RD criterion and qualitative image quality. Furthermore, we confirmed by visualizing our network parameters that the NN model was appropriately overfitted to represent the edges and textures precisely. Although we used empirical values for $\lambda_{y'}$ and $\lambda_{\psi'}$, these parameters should be optimized for each target image. Additionally, a more powerful compression method for ψ' is required to achieve a better RD performance. These two points will be addressed in future work.

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